Scenario-Cover Geometry for a Four-Terminal Planar Unsplittable-Flow Gadget

Integrated technical synopsis with companion proofs

Author line intentionally withheld pending final approval

2 August 2026 – private RC1 human-review repair, not released

Scope and status. This technical synopsis integrates a fixed-gadget theorem program supported by versioned evidence and complete companion proofs in the candidate repository. It is not an unrestricted planar-SSUF theorem, complete external human review, peer review, proof-assistant verification, novelty clearance, or release authority. The controlling one-scenario, repaired high-κ, and frozen review sources have been authenticated. R4 v6 accepted the cross-package scope statements, not the act of publication. One supplied external human report reconstructed the duality and short GM-005/006 arguments but did not inspect the full SC-006 or GM-008/009 proofs; it is scope-limited, not full acceptance. This synopsis is not a standalone substitute for its companion proof artifacts.

Abstract

We study a fixed planar acyclic single-source unsplittable-flow gadget with four terminals and two designated source–terminal paths per terminal. For strictly positive full-demand route-cost-difference scenarios, the following fixed-gadget scenario-count ladder is supported by complete analytic proof sources and narrower exact corroboration:

$$\beta_G^{(m,+)} = \begin{cases} (299-41\sqrt{41})/32, & m=1, \\ 17/8, & m=2, \\ 3, & m=3, \\ 4, & m\geq 4. \end{cases}$$

The two-, three-, and four-or-more-scenario suprema are not attained by finite legal instances. No attainment or nonattainment assertion is made for the one-scenario supremum. The one-scenario value also holds for the broader class of legally realizable signed and zero route-cost differences, but no such multi-scenario extension is claimed.

The $m = 1$ equality in this integrated ladder is the later private GM-002/003 arbitrary/legal fixed-gadget theorem. The immutable public v0.1.0/v0.2.1 one-scenario disclosure proves the same radical for its narrower lower family; the two statements must not be conflated.

The organizing principle is exact scenario-cover duality: to force all common cost-nonincreasing routings above a deviation threshold, the scenarios' open homogeneous halfspaces must cover every lower-deviation routing displacement. At one rational RB witness, an exact atlas classifies all 168 downsets of the four-cube, 59 realizable positive-scenario losing patterns, 32 condition-number infima, and every one- through four-scenario cover count. A separately isolated blind implementation reproduced the fixed finite mathematical data after lossless convention normalization; strict certificate payloads were not identical, and that distinction is retained.

For the RB family $p_\varepsilon = (1/4+\varepsilon, \varepsilon, 1/2, 1/4+\varepsilon)$, $d = (1, 1, 3/4, 1)$, we give an exact four-phase two-scenario condition-number diagram. The repaired global fixed-gadget theorem gives $\beta_G^{(2,+)}(\kappa) \leq 2$ for $1 \leq \kappa \leq 2$; for $\kappa > 2$ it gives $\beta_G^{(2,+)}(\kappa) \leq \max\{2, F(\kappa)\}$, with exact equality to $F(\kappa)$ for $\kappa \geq \kappa_0$, where κ_0 is the largest real root of an explicit quartic. The exact middle curve below κ_0 remains open.

Contents

1 The fixed gadget and robust objective

The graph is the released four-terminal planar DAG. It has a five-arc trunk $a_1, \ldots, a_5$, and each terminal $i \in [4]$ has an E path and a C path. Relative to the E path, switching terminal i to its C path adds its full demand on the trunk support

$$I_1 = \{a_1, a_2, a_3\}, \quad I_2 = \{a_1, a_2, a_3, a_4, a_5\}, \quad I_3 = \{a_2, a_3, a_4, a_5\}, \quad I_4 = \{a_3, a_4\}. \tag{1.1}$$

There are also two private arcs per terminal, so every candidate routing is checked on thirteen arcs.

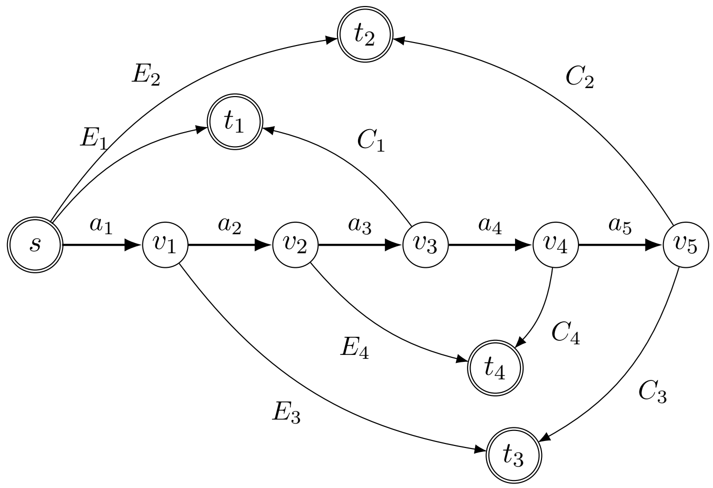

Figure 1: Schematic of the released two-path gadget. The proof uses the path-difference supports in (1.1); private arcs permit legal realization of full-demand route-cost differences.

Let $0 < d_i \leq 1$ and normalize $d_{\max} = 1$. Let $p_i \in [0, 1]$ be the fractional C fraction and let $S \subseteq [4]$ be the C-set of an unsplittable routing. Write

$$v_S := \mathbf{1}_S - p \in \mathbb{R}^4. \tag{1.2}$$

For a full-demand E-minus-C route-cost-difference vector k,

$$c(y^S) - c(x) = k \cdot p - k(S) = -k \cdot v_S. \tag{1.3}$$

Thus S is cost-nonincreasing exactly when

$$k \cdot v_S \geq 0. \tag{1.4}$$

Every positive k is physically realizable using nonnegative, commodity-independent arc costs: place cost k_i/d_i on an E-private arc of terminal i and zero on the corresponding C-private arc and all other arcs.

Let $M(S; p, d)$ be the maximum positive deviation of route S from the fractional flow over all thirteen arcs. For a trunk arc a, direct path-incidence subtraction gives

$$\Delta_a(S) = \sum_{\substack{i \in S \\ a \in I_i}} h_i - \sum_{\substack{i \notin S \\ a \in I_i}} \ell_i, \qquad h_i = d_i(1 - p_i), \quad \ell_i = d_i p_i, \tag{1.4a}$$

where S is the C-set. Thus

$$M(S; p, d) \leq \max\left\{1, \sum_{i \in S} h_i\right\} \leq \max\{1, |S|\}. \tag{1.4b}$$

The first term covers all eight terminal-private arcs: each positive private deviation is either h_i or ℓ_i, hence at most $d_i \leq 1$. Equations (1.4a)–(1.4b) are proved without enumeration in the companion `research/fixed_gadget_scenario_cover/scenario_ladder/TRUNK_PRIVATE_ARC_ENVELOPE.md` and structurally checked for all sixteen routings by its exact verifier.

For positive scenarios $k^{(1)}, \dots, k^{(m)}$, define

$$\Phi(p, d; k^{(1)}, \dots, k^{(m)}) := \min_{\substack{S \subseteq [4] \\ k^{(j)} \cdot v_S \geq 0 \ \forall j}} M(S; p, d). \tag{1.5}$$

The feasible family is nonempty: the all-C C-set [4] has displacement $\mathbf{1} - p \geq 0$ and is feasible in every positive scenario. Let $\beta_G^{(m,+)}$ be the supremum of (1.5) over legal normalized data on this fixed gadget.

Some proofs are shorter in the E-set orientation $R = [4] \setminus S$. Put $q = \mathbf{1} - p$. Then R is feasible in a positive scenario exactly when

$$k(R) \leq k \cdot q. \tag{1.6}$$

2 Scenario-cover duality

For one positive scenario define the eliminated downset

$$\mathcal{B}(k; p) := \{S \subseteq [4] : k \cdot v_S < 0\}. \tag{2.1}$$

For a target deviation t, define

$$\mathcal{G}_{<t}(p, d) := \{S \subseteq [4] : M(S; p, d) < t\}. \tag{2.2}$$

Strictness is essential: a route tying a scenario budget is feasible, and a route tying t need not be eliminated to force value at least t.

Theorem 2.1 (Exact scenario-cover duality). *For positive scenarios $k^{(1)}, \dots, k^{(m)}$,*

$$\Phi(p, d; k^{(1)}, \dots, k^{(m)}) \geq t \quad \Longleftrightarrow \quad \mathcal{G}_{<t}(p, d) \subseteq \bigcup_{j=1}^{m} \mathcal{B}(k^{(j)}; p). \tag{2.3}$$

Proof. The left side says that no common feasible routing has deviation below t. By (1.4), a route is not common feasible exactly when at least one scenario has negative dot product with its displacement. This is precisely the cover statement on the right. □

For $\kappa \geq 1$, let

$$\mathcal{K}_\kappa := \left\{ k \in \mathbb{R}^4_{>0} : \frac{\max_i k_i}{\min_i k_i} \leq \kappa \right\}. \tag{2.4}$$

For a route family $\mathcal{A}$, define

$$\tau_{\kappa,p}(\mathcal{A}) := \min \left\{ m : \mathcal{A} \subseteq \bigcup_{j=1}^{m} \mathcal{B}(k^{(j)}; p),\ k^{(j)} \in \mathcal{K}_\kappa \right\}. \tag{2.5}$$

Use $\tau_{\kappa,p}(\varnothing) = 0$ and value $+\infty$ when no cover exists; $\kappa = \infty$ means unrestricted positive normals. A cover with fewer than m scenarios may be padded by duplicating a positive scenario, so exact and at-most conventions agree for the value. Cover thresholds are infima and strict endpoint availability is checked separately. If $\mathcal{R}(p, d)$ is the finite set of route values,

$$\Psi_{m,\kappa}(p, d) = \max \left\{ r \in \mathcal{R}(p, d) : \tau_{\kappa,p}(\mathcal{G}_{<r}(p, d)) \leq m \right\}. \tag{2.6}$$

Locally, $\beta_G^{(m,+)}(\kappa)$ denotes the supremum of (2.6) over $p \in [0, 1]^4$ and positive normalized d on this fixed gadget. Multi-scenario robust rounding is therefore a restricted open-halfspace cover problem on sixteen routing displacement vectors.

3 The fixed-gadget scenario-count ladder

Let

$$L := \frac{299 - 41\sqrt{41}}{32} = 1.139747070789\ldots. \tag{3.1}$$

The integrated private claim ledger is

$$\boxed{\beta_G^{(m,+)} = \begin{cases} L, & m = 1, \\ 17/8, & m = 2, \\ 3, & m = 3, \\ 4, & m \geq 4. \end{cases}} \tag{3.2}$$

The evidence is modular. R1 v2 accepted GM-002 and GM-003 at claim level while its packet retained editorial repairs; the complete derived one-scenario proof and pinned dependencies are companion artifacts under `research/fixed_gadget_scenario_cover/one_scenario/`. The $m = 2$ proof remains the immutable unrefereed v0.2.1 RB-003 package. R2 v2 accepted the repaired GM-005 proof, and original R2 accepted GM-006 at claim level. The complete current GM-005 and GM-006 proofs are companion artifacts under `scenario_ladder/`; the short arguments below are their synopsis. The complete, hash-bound reviewer index is `research/fixed_gadget_scenario_cover/FULL_PROOF_REVIEW_MAP.md`.

3.1 Three positive scenarios

Use the E-set orientation and put

$$h_i = d_i q_i, \qquad H = \sum_i h_i. \tag{3.3}$$

The all-C route is always feasible and has exact value H.

Lemma 3.1 (Singleton assignment)**.** *If every singleton E-set is blocked by at least one of three positive scenarios, then $H < 3$.*

Proof. If $q = 0$, then $H = 0 < 3$. Otherwise set $B_j = k^{(j)} \cdot q > 0$ and $w^{(j)} = k^{(j)}/B_j$, so $w^{(j)} \cdot q = 1$. Assign each singleton to one scenario that blocks it, with assigned group G_j. Blocking is strict, so $w_i^{(j)} > 1$ for $i \in G_j$. For a positive-mass group,

$$\sum_{i \in G_j} h_i < \sum_{i \in G_j} h_i w_i^{(j)} \leq \sum_i q_i w_i^{(j)} = 1, \tag{3.4}$$

because $h_i = d_i q_i \leq q_i$. Empty or zero-mass groups contribute $0 < 1$. Summing gives $H < 3$. □

Lemma 3.2 (Equality collapse)**.** *If some singleton E-set $\{r\}$ is feasible, the instance has a feasible routing of value strictly below three.*

Proof. The singleton route has at most three positive C-terminal contributions on any trunk, each at most one, and private deviation at most one. Its value is therefore at most three. Equality forces $h_i = d_i q_i = 1$ for every $i \neq r$, hence $d_i = q_i = 1$ for those three terminals. The complementary E-triple $T = [4] \setminus \{r\}$ is feasible in every scenario because

$$k^{(j)}(T) = \sum_{i \neq r} k_i^{(j)} = \sum_{i \neq r} q_i k_i^{(j)} \leq k^{(j)} \cdot q. \tag{3.5}$$

In route T, the three E-routed terminals have $d_i(1 - q_i) = 0$; only terminal r can contribute positively, so the route has value at most one. □

Theorem 3.3 (Three-scenario branch)**.** $\beta_G^{(3,+)} = 3$*, and every finite legal instance has value strictly below three.*

Proof. Either all singletons are blocked and lemma 3.1 gives the all-C route value $H < 3$, or a singleton is feasible and lemma 3.2 gives a route below three. For the lower sequence, take an integer $n \geq 2$, unit demands, $q_i = 1 - 1/n$, and

$$(3n, 1, 1, 1), \quad (1, 3n, 1, 1), \quad (1, 1, 3n, 1). \tag{3.6}$$

The common feasible E-sets are exactly $\varnothing$ and $\{4\}$. The latter route has exact value $3(1-1/n) \to 3$. $\square$

3.2 Four or more positive scenarios

Theorem 3.4 (Many-scenario branch)**.** *For every integer $m \geq 4$, $\beta_G^{(m,+)} = 4$, and every finite legal instance has value strictly below four.*

Proof. The all-C route has value $H = \sum_i d_i q_i \leq 4$. For an integer $n \geq 2$, unit demands, and $q_i = 1-1/n$, use

$$(3n, 1, 1, 1), \quad (1, 3n, 1, 1), \quad (1, 1, 3n, 1), \quad (1, 1, 1, 3n). \tag{3.7}$$

Every nonempty E-set is blocked by a scenario heavy on one of its terminals, so only $\varnothing$ is common feasible, with value $4(1 - 1/n) \to 4$. Repeat a scenario for $m > 4$.

If a finite instance had value four, the all-C route would force $d_i = q_i = 1$ for all i. The full E-set would then be weakly feasible at equality in every scenario and would have zero deviation, a contradiction. $\square$

Remark 3.5. The $m = 2$, $m = 3$, and $m \geq 4$ values are nonattained. GM-002 is a supremum-only statement; this synopsis asserts neither attainment nor nonattainment for $m = 1$.

4 Exact finite scenario-cover atlas

At the rational RB witness

$$p = \left(\frac{251}{1000}, \frac{1}{1000}, \frac{1}{2}, \frac{251}{1000}\right), \qquad d = \left(1, 1, \frac{3}{4}, 1\right), \tag{4.1}$$

the sixteen route levels are as follows.

The exact atlas classifies all 168 downsets of the four-cube. Exactly 59 are one-scenario losing families at this p, with 32 distinct condition-number infima. Minimum cover counts are

cover count	0	1	2	3	4	impossible
downsets	1	61	91	13	1	1.

(4.2)

The unique four-scenario downset is all proper C-sets; the full cube is impossible because all-C cannot be eliminated by a positive scenario.

With unrestricted positive normals, the fixed point has

m	1	2	3	4
$\Psi_{m,\infty}(p, d)$	561/500	1061/500	2123/1000	359/125.

(4.3)

These are fixed-instance encoded values, not the global fixed-gadget values in (3.2). The checkers do not certify unencoded orientation, analytic case exhaustion, global optimization, dependency correctness, statement scope, novelty, authorship, or rights.

Table 1: Exact route values at the finite RB witness.

C-set(s)	$M(S;p,d)$
$\varnothing, 3$	$3/8$
$1, 4$	$749/1000$
2	$999/1000$
14	$561/500$
$13, 34$	$1123/1000$
24	$1373/1000$
23	$687/500$
12	$437/250$
134	$234/125$
124	$1061/500$
$123, 234$	$2123/1000$
1234	$359/125$

For two scenarios, set

$$A = \frac{1498}{501}, \qquad B = 998, \qquad C = 1998. \tag{4.4}$$

Then

$$\Psi_{2,\kappa}(p,d) = \begin{cases} 561/500, & 1 \le \kappa \le A, \\ 1123/1000, & A < \kappa \le B, \\ 234/125, & B < \kappa \le C, \\ 1061/500, & C < \kappa. \end{cases} \tag{4.5}$$

Each new phase starts strictly after its breakpoint.

4.1 Blind finite reconstruction

R3B v2 was developed under technically enforced answer isolation, frozen before expected outputs and comparison code were disclosed, and returned `ACCEPT_AS_STATED`. For all 451 compared certificates, the primal and objective magnitude agreed, and both certificate populations exact-validated after lossless convention projection. Fourteen degenerate LPs used alternative valid duals.

The raw strict-payload comparator nevertheless returned `FAIL`. The correct label is

$$\boxed{\texttt{SEMANTIC\_MATHEMATICAL\_MATCH / STRICT\_CERTIFICATE\_PAYLOAD\_IDENTITY\_FAIL.}} \tag{4.6}$$

Thus `BLIND_INDEPENDENT_REPRODUCTION` applies only to this fixed finite atlas.

5 Exact RB-family two-scenario phase

For $0 < \varepsilon < 1/8$, set

$$p_\varepsilon = \left(\frac{1}{4} + \varepsilon, \varepsilon, \frac{1}{2}, \frac{1}{4} + \varepsilon\right), \qquad d = \left(1, 1, \frac{3}{4}, 1\right), \tag{5.1}$$

and

$$A(\varepsilon) = \frac{3 - 4\varepsilon}{1 + 2\varepsilon}, \qquad B(\varepsilon) = \frac{1}{\varepsilon} - 2, \qquad C(\varepsilon) = \frac{2}{\varepsilon} - 2. \tag{5.2}$$

Put

$$r_0 = \frac{9}{8} - 3\varepsilon, \quad r_1 = \frac{9}{8} - 2\varepsilon, \quad r_2 = \frac{15}{8} - 3\varepsilon, \quad r_3 = \frac{17}{8} - 3\varepsilon. \tag{5.3}$$

Theorem 5.1 (SC-006)**.** *For exactly two positive scenarios of condition number at most κ,*

$$\Psi_{2,\kappa}(p_\varepsilon, d) = \begin{cases} r_0, & 1 \le \kappa \le A(\varepsilon), \\ r_1, & A(\varepsilon) < \kappa \le B(\varepsilon), \\ r_2, & B(\varepsilon) < \kappa \le C(\varepsilon), \\ r_3, & C(\varepsilon) < \kappa. \end{cases} \tag{5.4}$$

Every breakpoint belongs to the lower branch.

The first transition is controlled by route 14:

$$T(1, \kappa, \kappa, 1) - k(14) = \left(\frac{1}{2} + \varepsilon\right)(\kappa - A(\varepsilon)), \tag{5.5}$$

with a matching upper inequality for every κ-bounded scenario.

For the middle transition, if one scenario eliminates any two of the outer pairs $13, 14, 34$, adding the strict losing inequalities gives

$$\mathrm{cond}(k) > \frac{1 - 2\varepsilon}{\varepsilon} = B(\varepsilon). \tag{5.6}$$

Two scenarios covering three outer pairs therefore force one across B. Above B, the star–triangle normals

$$(1, \kappa, 1, 1), \qquad (\kappa, 1, \kappa, \kappa) \tag{5.7}$$

cover all six pairs.

The third transition is controlled by route 134:

$$T(k) - k(134) \le \varepsilon(\kappa - C(\varepsilon)) \min_i k_i. \tag{5.8}$$

Above C, the first normal in (5.7) also eliminates 134.

Finally, one scenario cannot eliminate both triples 134 and 124. A scenario eliminating 134 accepts every set containing terminal 2, while a scenario eliminating 124 accepts every set containing terminal 3. Two scenarios assigned one triple each therefore both accept 23. This proves the unrestricted fixed-family ceiling r_3.

The endpoint $\varepsilon = 1/8$ is excluded because the route filtration changes. SC-006 is fixed-family, exactly-two-scenario, and positive-coordinate only. The complete reviewed proof is the companion artifact `scenario_cover/SC006_CONTINUUM_THEOREM.md`; this section is a synopsis rather than a replacement for that proof.

6 High-heterogeneity global tail

Define

$$F(\kappa) = \frac{17\kappa^4 - 22\kappa^3 - 13\kappa^2 + 18\kappa + 1}{8(\kappa^2 - 1)^2}, \tag{6.1}$$

and let κ_0 be the largest real root of

$$P(\kappa) = \kappa^4 - 22\kappa^3 + 19\kappa^2 + 18\kappa - 15. \tag{6.2}$$

Then

$$\kappa_0 = 21.058780922898283\ldots, \qquad F(\kappa_0) = 2. \tag{6.3}$$

Theorem 6.1 (Internally reviewed high-κ tail and global bounds)**.** *For the fixed gadget and exactly two positive scenarios,*

$$L \le \beta_G^{(2,+)}(\kappa) \le 2 \qquad (1 \le \kappa \le 2),$$

$$\max\{L, F(\kappa)\} \le \beta_G^{(2,+)}(\kappa) \le 2 \qquad (2 < \kappa < \kappa_0),$$

and

$$\boxed{\beta_G^{(2,+)}(\kappa) = F(\kappa) \qquad (\kappa \ge \kappa_0).} \tag{6.4}$$

The cover geometry explains the parameters. If Q is the central E fraction and S the total outer E fraction, the center-heavy and outer-heavy cover boundaries are

$$S = \kappa(1 - Q), \qquad S = 2 - \frac{Q}{\kappa}. \tag{6.5}$$

Their intersection is

$$Q_\kappa = \frac{\kappa(\kappa - 2)}{\kappa^2 - 1}, \qquad S_\kappa = \frac{\kappa(2\kappa - 1)}{\kappa^2 - 1}. \tag{6.6}$$

Substitution into the accepted load envelope gives $F(\kappa) = Q_\kappa + (1 + S_\kappa)^2/8$, and

$$F(\kappa) - 2 = \frac{P(\kappa)}{8(\kappa - 1)^2(\kappa + 1)^2}. \tag{6.7}$$

The complete companion proof is `bounded_heterogeneity/GM008_GM009_HIGH_KAPPA_THEOREM.md`. It contains the RB-003 value-above-two branch reduction, strict star and triangle inequalities, the early $t \le 2$ split, exact three-regime allocation envelope, one-variable maximization, strict lower family, and quartic-root argument. In particular, a feasible outer singleton first gives $t \le \max\{1, H - h_i\}$; only the standing $t > 2$ branch yields $t \le H - h_i$. The exact curve on $1 \le \kappa < \kappa_0$ remains open; the upper bound by two is not an equality classification. No claim of finite attainment for $F(\kappa)$ is made here.

7 Related and concurrent work

The primary records were checked on 2 August 2026. Sergey Nikolenko's first manuscript is dated 29 July 2026 and its Zenodo record was published 30 July 2026 (DOI 10.5281/zenodo.21701162). It presents the same radical L through the same four crossing interval supports. His second manuscript and Zenodo record are dated 31 July 2026 (DOI 10.5281/zenodo.21716713) and give a seventeen-terminal common-point interval construction with exact lower bound

$$\frac{1282494797984843521}{10^{18}} = 1.282494797984843521\ldots, \tag{7.1}$$

with an exhaustive 2^{17}-routing check and an eighteen-atom exact convex-hull certificate. It is therefore the stronger public numerical lower bound for the unrestricted ordinary one-scenario planar problem among these records. It does not resolve the fixed-gadget multi-scenario constants, scenario-cover atlas, SC-006, or high-κ tail studied here.

The repository's immutable first release is dated 23 July 2026. These dates are recorded for neutral chronology only. This draft makes no novelty, priority, dependence, independent-discovery, or misconduct inference. The two Zenodo entries' titles, dates, DOI metadata, and stated headline bounds have been audited; broader literature coverage, overlap analysis, citations, and acknowledgements still require neutral review.

8 Complete proof map and computation boundary

Every controlling analytic source is included in the candidate. Revised reviewer packages expose these as ordinary unpacked, line-numbered files rather than leaving them only inside a nested source archive. The hash-bound index is `research/fixed_gadget_scenario_cover/FULL_PROOF_REVIEW_MAP.md`. The controlling sources are:

- **GM-002/003:** `GLOBAL_ARBITRARY_ONE_SCENARIO_THEOREM.md`, in the `one_scenario/proof/` directory, with the pinned baseline proofs.
- **GM-004:** `TWO_SCENARIO_GLOBAL_CONSTANT.md`, the immutable public RB-003 proof.
- **GM-005/006:** `GM005_THREE_SCENARIO_THEOREM.md`, `GM006_FOUR_OR_MORE_SCENARIOS.md`, and the standalone thirteen-arc envelope in the `scenario_ladder/` directory.
- **SC-006:** `SC006_CONTINUUM_THEOREM.md`, containing the analytic obstructions, constructions, endpoint analysis, and global RB-family ceiling.
- **GM-008/009:** `GM008_GM009_HIGH_KAPPA_THEOREM.md`, containing the RB-003 branch reduction, strict cover inequalities, allocation envelope, one-variable maximization, matching lower family, and quartic analysis.

The exact finite atlas is authoritative only for its stated fixed finite instance. The SC-006 and high-κ equalities rest on the analytic sources above; enumeration, symbolic scripts, exact samples, mutation tests, and replay are corroboration and regression evidence, not substitutes for global case analysis.

9 Evidence hierarchy and release gates

The evidence hierarchy is:

1. human-readable theorem proofs and exact derivations;
2. exact finite certificates when the claim itself is finite;
3. independently authored or isolated reconstruction within its declared scope;
4. deterministic replay, mutations, and symbolic corroboration; and
5. numerical reconnaissance, used only to suggest questions.

The frozen R1–R4 review lanes are internal and AI-assisted. One external human report was supplied on 2 August 2026; the supplied text does not identify the reviewer or document independence/conflicts. It reconstructed duality and the short GM-005/006 arguments, reported no mathematical contradiction in the portions examined, and did not inspect the full SC-006 or GM-008/009 companion proofs. It is therefore scope-limited external criticism, not a complete claim-by-claim review, journal peer review, or formal proof-assistant verification.

The controlling proof sources are authenticated, and frozen R4 v6 accepted all 23 statement-and-scope integration rows without a finding. That does not open a public gate.

Before any public action:

- preserve the completed two-clean-root RC1 build and source/PDF audit;
- finish the broader neutral literature and overlap review;

- obtain scoped independent human reconstruction of the controlling proofs, especially SC-006 and GM-008/009;
- settle authorship, acknowledgements, and licensing; and
- obtain explicit approval for the exact diff and communication.

10 Open mathematical problems

1. Determine the global fixed-gadget two-scenario curve for $1 \leq \kappa < \kappa_0$.
2. Prove or refute the exact F064 value $5/2 - \sqrt{2}$; currently only a strict lower sequence is proved.
3. Classify marginal thresholds across all four-interval support topologies and search higher-terminal gadgets using separation-generated costs.
4. Study restricted scenario-cover numbers under topology changes, terminal cloning, and common-point interval systems.
5. Formalize the finite atlas and shortest analytic components in a proof assistant, without confusing formalization with external peer review.

11 Claim and nonclaim ledger

ID	Statement	Staging posture
GM-002	One legal scenario has supremum L, including legally realizable signed and zero differences; supremum only.	R1 v2 claim `ACCEPT_AS_STATED`; packet editorial; source authenticated.
GM-004	Two positive scenarios have supremum 17/8, nonattained.	Public unrefereed v0.2.1 theorem.
GM-005	Three positive scenarios have supremum 3, nonattained.	Repaired R2 v2 `ACCEPT_AS_STATED`; original R2 history preserved.
GM-006	Four or more positive scenarios have supremum 4, nonattained.	Original-R2 claim-level `ACCEPT_AS_STATED`; derived proof included.
SC-000	Scenario forcing equals an open-halfspace cover of routes below the target.	Internal R3A acceptance with nonblocking repairs applied here.
SC-001–003	Complete 168-downset atlas at the RB witness.	Exact finite result; narrow blind reconstruction.
SC-006	Exact four-phase two-scenario curve on the fixed RB family.	Internally AI-reviewed narrow theorem; nonblind replay.
GM-008/009	Global upper bound and $F(\kappa)$ tail for $\kappa \geq \kappa_0$.	Repaired R2 v3 claims and packet `ACCEPT_AS_STATED`; two nonblocking findings retained.
GM-010	Global bounded-κ curve below κ_0.	Open.
GM-014	Exact F064 value $5/2 - \sqrt{2}$.	Unclaimed; lower sequence only.

ID	Statement	Staging posture
GM-015	Topology-wide or unrestricted planar sharpness.	Explicit nonclaim.

Acknowledgement and rights placeholder

The project was developed through a human-directed, AI-assisted research process involving multiple model sessions and code/review lanes. A public version must identify the human author, contribution and stewardship roles, acknowledgements, and rights language deliberately. Repository ownership does not by itself settle authorship or legal ownership. This RC1 candidate contains no license grant.

The Nikolenko entries below have passed a primary-record metadata and headline claim check. The bibliography as a whole remains provisional and has not passed the broader neutral citation, chronology, or overlap audit required before publication.

References

[1] Y. Dinitz, N. Garg, and M. X. Goemans, *On the single-source unsplittable flow problem*, Combinatorica 19 (1999), 17–41.

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